Week 6: Product Review and Rating Management System

1. Create Review and Rating Tables

```
SQL> CREATE TABLE Review(ReviewID VARCHAR2(10), CustomerID VARCHAR2(20),
 ProductID NUMBER(5), Rating NUMBER(1), ReviewTest VARCHAR2(50), ReviewD
ate DATE, PRIMARY KEY(ReviewID), FOREIGN KEY(CustomerID) REFERENCES Cust
omer(CUSTOMER_ID), FOREIGN KEY(ProductID) REFERENCES Product(PRODUCT_ID)
);

Table created.

SQL> desc Review
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------
--------
 REVIEWID                                  NOT NULL VARCHAR2(10)
 CUSTOMERID                                         VARCHAR2(20)
 PRODUCTID                                          NUMBER(5)
 RATING                                             NUMBER(1)
 REVIEWTEST                                         VARCHAR2(50)
 REVIEWDATE                                         DATE
```

Store Customer Feedback and Ratings Tasks:

1. Add reviews for purchased products.

```
SQL> INSERT INTO REVIEW VALUES('REV0001','CUST000001',101,5,'Excellent moisturizer',DATE '2026-01-20');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0002','CUST000002',102,4,'Good face wash',DATE '2026-01-21');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0003','CUST000003',103,5,'Highly effective sunscreen',DATE '2026-02-02');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0004','CUST000004',104,4,'Very good serum',DATE '2026-02-03');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0005','CUST000005',106,5,'Excellent herbal moisturizer',DATE '2026-02-15');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0006','CUST000006',107,4,'Good moisturizing effect',DATE '2026-02-18');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0007','CUST000007',108,5,'Amazing hydration',DATE '2026-02-20');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0008','CUST000008',109,3,'Average product quality',DATE '2026-02-22');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0009','CUST000009',110,4,'Nice glow effect',DATE '2026-02-25');
1 row created.
SQL> INSERT INTO REVIEW VALUES('REV0010','CUST000010',101,5,'Best moisturizer I have used',DATE '2026-02-28');
1 row created.
```

```
SQL> SELECT * FROM REVIEW;

REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                            REVIEWDAT
----------------------------------------------------- ---------
REV0001    CUST000001                  101          5
Excellent moisturizer                                 20-JAN-26

REV0002    CUST000002                  102          4
Good face wash                                        21-JAN-26

REV0003    CUST000003                  103          5
Highly effective sunscreen                            02-FEB-26


REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                            REVIEWDAT
----------------------------------------------------- ---------
REV0004    CUST000004                  104          4
Very good serum                                       03-FEB-26

REV0005    CUST000005                  106          5
Excellent herbal moisturizer                          15-FEB-26

REV0006    CUST000006                  107          4
Good moisturizing effect                              18-FEB-26


REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                            REVIEWDAT
----------------------------------------------------- ---------
REV0007    CUST000007                  108          5
Amazing hydration                                     20-FEB-26

REV0008    CUST000008                  109          3
Average product quality                               22-FEB-26

REV0009    CUST000009                  110          4
Nice glow effect                                      25-FEB-26


REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                            REVIEWDAT
----------------------------------------------------- ---------
REV0010    CUST000010                  101          5
Best moisturizer I have used                          28-FEB-26


10 rows selected.
```

2. Store customer ratings from 1 to 5 stars.

Ensure this by adding Constraint

```
SQL> ALTER TABLE REVIEW ADD CONSTRAINT CHK_RATE CHECK(RATING >= 1 AND RATING <= 5);

Table altered.

SQL> INSERT INTO REVIEW VALUES('REV0019','CUST000001',101,6,'Excellent product',SYSDATE);
INSERT INTO REVIEW VALUES('REV0019','CUST000001',101,6,'Excellent product',SYSDATE)
*
ERROR at line 1:
ORA-02290: check constraint (C##THIRU.CHK_RATE) violated
```

3. Update review comments if required.

```
SQL> UPDATE REVIEW SET REVIEWTEST='Good quality but could be improved' WHERE REVIEWID='REV0010';

1 row updated.
```

4. Remove inappropriate or invalid reviews.

```
SQL> DELETE FROM REVIEW WHERE REVIEWID='REV0009';

1 row deleted.
```

4. Retrieve Product Review Details

Write SQL queries to:

1. Display all reviews for a product.

```
SQL> SELECT * FROM REVIEW WHERE PRODUCTID=101;

REVIEWID   CUSTOMERID            PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                            REVIEWDAT
----------------------------------------------------- ---------
REV0001    CUST000001                  101          5
Excellent moisturizer                                 20-JAN-26

REV0010    CUST000010                  101          5
Good quality but could be improved                    28-FEB-26
```

2. Display customer name with their review

```
SQL> SELECT CONCAT(c.First_Name,c.Last_Name) AS Customer_Name, R.ReviewID, R.ProductID, R.Rating, R.ReviewTest FROM Customer C JOIN Review R ON C.Customer_ID = R.CustomerID
CUSTOMER_NAME                  REVIEWID   PRODUCTID    RATING
------------------------------ ---------- ---------- ----------
REVIEWTEST
--------------------------------------------------
ThiruR                         REV0001         101          5
Excellent moisturizer

AnuK                           REV0002         102          4
Good face wash

PriyaS                         REV0003         103          5
Highly effective sunscreen

CUSTOMER_NAME                  REVIEWID   PRODUCTID    RATING
------------------------------ ---------- ---------- ----------
REVIEWTEST
--------------------------------------------------
KaviyaM                        REV0004         104          4
Very good serum

DivyaP                         REV0005         106          5
Excellent herbal moisturizer

NishaS                         REV0006         107          4
Good moisturizing effect

CUSTOMER_NAME                  REVIEWID   PRODUCTID    RATING
------------------------------ ---------- ---------- ----------
REVIEWTEST
--------------------------------------------------
KeerthiR                       REV0007         108          5
Amazing hydration

MeenaT                         REV0008         109          3
Average product quality

AarthiK                        REV0010         101          5
Good quality but could be improved

9 rows selected.
```

3. Find products having maximum reviews.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, COUNT(R.REVIEWID) AS REVIEW_COUNT FROM PRODUCT P JOIN REVIEW R ON
P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME ORDER BY REVIEW_COUNT DESC;

PRODUCT_ID PRODUCT_NAME                                       REVIEW_COUNT
---------- -------------------------------------------------- ------------
       101 Aloe Moisturizer                                              2
       109 Daily Moisturizer                                             1
       108 Hydra Moisturizer                                             1
       104 Vitamin C Serum                                               1
       107 Vitamin Moisturizer                                           1
       106 Herbal Moisturizer                                            1
       102 Gentle Face Wash                                              1
       103 SPF 50 Sunscreen                                              1

8 rows selected.
```

4. Display recent customer feedback.

```
SQL> SELECT * FROM (SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME, P.PRODUCT_NAME, R.RATING, R.
REVIEWTEST AS CUSTOMER_FEEDBACK, R.REVIEWDATE AS REVIEW_DATE FROM REVIEW R JOIN CUSTOMER C ON R.CUSTOMERID =
 C.CUSTOMER_ID JOIN PRODUCT P ON R.PRODUCTID = P.PRODUCT_ID ORDER BY R.REVIEWDATE DESC) WHERE ROWNUM <= 5;

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------
Aarthi K
Aloe Moisturizer                                            5
Good quality but could be improved                 28-FEB-26

Meena T
Daily Moisturizer                                           3
Average product quality                            22-FEB-26

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------

Keerthi R
Hydra Moisturizer                                           5
Amazing hydration                                  20-FEB-26

Nisha S
Vitamin Moisturizer                                         4

CUSTOMER_NAME
----------------------------------------
PRODUCT_NAME                                       RATING
-------------------------------------------------- ----------
CUSTOMER_FEEDBACK                                  REVIEW_DA
-------------------------------------------------- ---------
Good moisturizing effect                           18-FEB-26

Divya P
Herbal Moisturizer                                          5
Excellent herbal moisturizer                       15-FEB-26
```

5. Retrieve reviews with ratings above 4.

```
SQL> SELECT * FROM Review WHERE Rating > 4;

REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0001    CUST000001                  101          5
Excellent moisturizer                               20-JAN-26

REV0003    CUST000003                  103          5
Highly effective sunscreen                          02-FEB-26

REV0005    CUST000005                  106          5
Excellent herbal moisturizer                        15-FEB-26


REVIEWID   CUSTOMERID           PRODUCTID     RATING
---------- -------------------- ---------- ----------
REVIEWTEST                                          REVIEWDAT
--------------------------------------------------- ---------
REV0007    CUST000007                  108          5
Amazing hydration                                   20-FEB-26

REV0010    CUST000010                  101          5
Good quality but could be improved                  28-FEB-26
```

Example Query:

5. Calculate Average Product Ratings

Use aggregate functions to analyze ratings.

Tasks:

1. Calculate average rating for each product.

```
SQL> SELECT P.PRODUCT_NAME, AVG(R.RATING) AS AVERAGE_RATING FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R
.PRODUCTID GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
SPF 50 Sunscreen                                                5
Vitamin Moisturizer                                             4
Herbal Moisturizer                                              5
Hydra Moisturizer                                               5
Aloe Moisturizer                                                5
Daily Moisturizer                                               3
Gentle Face Wash                                                4
Vitamin C Serum                                                 4

8 rows selected.
```

2. Count the number of reviews per product.

```
SQL> SELECT P.PRODUCT_NAME, COUNT(R.RATING) AS COUNT_RATING FROM PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R
.PRODUCTID GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME                                       COUNT_RATING
-------------------------------------------------- ------------
SPF 50 Sunscreen                                              1
Vitamin Moisturizer                                           1
Herbal Moisturizer                                            1
Hydra Moisturizer                                             1
Aloe Moisturizer                                              2
Daily Moisturizer                                             1
Gentle Face Wash                                              1
Vitamin C Serum                                               1

8 rows selected.
```

3. Find highest-rated products.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, ROUND(AVG(R.RATING),2) AS AVERAGE_RATING FROM PRODUCT P JOIN REVIE
W R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME HAVING AVG(R.RATING) = (SELECT MAX(A
VG_RATING) FROM (SELECT AVG(RATING) AS AVG_RATING FROM REVIEW GROUP BY PRODUCTID));

PRODUCT_ID PRODUCT_NAME                                        AVERAGE_RATING
---------- -------------------------------------------------- --------------
       108 Hydra Moisturizer                                               5
       103 SPF 50 Sunscreen                                                5
       106 Herbal Moisturizer                                              5
       101 Aloe Moisturizer                                                5
```

4. Identify products with average rating above 4.

```
SQL> SELECT P.PRODUCT_ID, P.PRODUCT_NAME, ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING FROM PRODUCT P JOIN REVI
EW R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME HAVING AVG(R.RATING) > 4 ORDER BY A
VERAGE_RATING DESC;

PRODUCT_ID PRODUCT_NAME                                        AVERAGE_RATING
---------- -------------------------------------------------- --------------
       108 Hydra Moisturizer                                               5
       101 Aloe Moisturizer                                                5
       106 Herbal Moisturizer                                              5
       103 SPF 50 Sunscreen                                                5
```

6. Identify Highly Rated Products

Generate product rating reports

. Report 1: Product Rating Analysis

Display: ● Product name ● Number of reviews ● Average rating

```
SQL> SELECT P.PRODUCT_NAME, COUNT(R.REVIEWID) AS NUMBER_OF_REVIEWS, ROUND(AVG(R.RATING), 2) AS AVERAGE_RATIN
G FROM PRODUCT P LEFT JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME ORDE
R BY AVERAGE_RATING DESC NULLS LAST;

PRODUCT_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
Hydra Moisturizer                                                  1
             5

SPF 50 Sunscreen                                                   1
             5

Herbal Moisturizer                                                 1
             5


PRODUCT_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
Aloe Moisturizer                                                   2
             5

Gentle Face Wash                                                   1
             4

Vitamin C Serum                                                    1
             4


PRODUCT_NAME                                       NUMBER_OF_REVIEWS
-------------------------------------------------- -----------------
AVERAGE_RATING
--------------
Vitamin Moisturizer                                                1
             4

Daily Moisturizer                                                  1
             3

Glow Moisturizer                                                   0


9 rows selected.
```

Report 2: Customer Feedback Analysis

Find: ● Most reviewed products

```
SQL> SELECT P.PRODUCT_NAME, COUNT(R.REVIEWID) AS REVIEW_COUNT FROM PRODUCT P JOIN REVIEW R ON P .PRODUCT_ID
= R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME ORDER BY REVIEW_COUNT DESC;

PRODUCT_NAME                                       REVIEW_COUNT
-------------------------------------------------- ------------
Aloe Moisturizer                                              2
Daily Moisturizer                                             1
Hydra Moisturizer                                             1
Vitamin C Serum                                               1
Vitamin Moisturizer                                           1
Herbal Moisturizer                                            1
Gentle Face Wash                                              1
SPF 50 Sunscreen                                              1

8 rows selected.
```

● Highly rated products

```
SQL> SELECT P.PRODUCT_NAME, ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING FROM PRODUCT P JOIN REVIEW R ON P.PROD
UCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME HAVING AVG(R.RATING) > 4 ORDER BY AVERAGE_RATING
DESC;

PRODUCT_NAME                                       AVERAGE_RATING
-------------------------------------------------- --------------
Hydra Moisturizer                                               5
Aloe Moisturizer                                                5
Herbal Moisturizer                                              5
SPF 50 Sunscreen                                                5
```

● Products requiring improvement

```
SQL> SELECT P.PRODUCT_NAME, COUNT(R.REVIEWID) AS REVIEW_COUNT, ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING FRO
M PRODUCT P JOIN REVIEW R ON P.PRODUCT_ID = R.PRODUCTID GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME HAVING AVG(R.R
ATING) < 3 ORDER BY AVERAGE_RATING;

no rows selected
```

Report 3: Rating Distribution

● Number of 5-star ratings

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATHINGS FROM REVIEW WHERE RATING = 5;

FIVE_STAR_RATHINGS
------------------
                 5
```

● Number of 4-star ratings

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATHINGS FROM REVIEW WHERE RATING = 4;

FOUR_STAR_RATHINGS
------------------
                 3
```

● Number of low-rated products

```
SQL> SELECT COUNT(*) AS LOW_RATED_PRODUCTS FROM (SELECT PRODUCTID FROM REVIEW GROUP BY PRODUCTID HAVING AVG(
RATING) < 3);

LOW_RATED_PRODUCTS
------------------
                 0
```

● Complete Rating

```
SQL> SELECT RATING, COUNT(*) AS NUMBER_OF_RATINGS FROM REVIEW GROUP BY RATING ORDER BY RATING DESC;

    RATING NUMBER_OF_RATINGS
---------- ------------------
         5                  5
         4                  3
         3                  1
```